



Nebulae supercomputer combines GPUs with CPUs

capabilities of CPUs but also greatly reduce the power consumption. Eight out of the world's greenest supercomputers combined specialised accelerators like GPUs with CPUs to boost performance and power efficiency, according to the Green500 list which is released twice a year.

Tianhe-1A, the latest supercomputer revealed in October 2010, parallels a large number of GPUs with multi-core CPUs to significantly boost its performance and reduce power consumption and size. In one of the releases, Guangming Liu, chief of National Supercomputer Center in Tianjin, said, "The performance and efficiency of Tianhe-1A was simply not possible without GPUs."

Other supercomputers combining GPUs with CPUs include the Dawning Nebulae and the Mole-8.5.

GGPGUs revolutionise co-processing

Co-processing with digital signal processor (DSP) blocks and floating-point units (FPUs) has enhanced the performance of CPUs. However, experts believe that a GPU co-processor will certainly mark a revolutionary step in the world of co-processing. GPUs have evolved more as general-purpose graphics processing unit (GGPGUs) mainly because of four reasons—massive parallelism, power, ubiquity and usability due to new tools and ease of development.

As parallel processors, GPUs excel

at tackling large amounts of similar data because the problem can be split into hundreds or thousands of pieces and calculated simultaneously. As sequential or serial processors, CPUs are not designed for this type of computation, but these are more adept at serial-based tasks such as running operating systems and organising data.

Goyal shares, "We believe that the concept of 'hybrid computing' or 'co-processing' where GPUs and CPUs work together will apply the most relevant processor to the specific task in hand, creating synergies."

Discrete vs integrated GPUs

As with any tool, it's very important to have the right GPU for the right task.

Graphics processors primarily exist in two form-factors, viz, discrete and integrated. Bhandari explains, "An integrated graphics processor means that when you buy a processor core system, the graphics component is part of the processor and you don't need to buy an additional piece of silicon or additional core to run the 3D graphics or high-end video. But, when you buy an additional core dedicated to graphics enhancement, it's called a discrete solution."

Explaining further, Goyal says, "If you are a gamer who demands the latest, greatest bleeding-edge entertainment experience with advanced features like stereoscopic 3D Vision, HD resolutions

and physical effects, you crave at least one high-end discrete GPU. But if you want to create a home media centre that is quiet and power-efficient, an integrated GPU may be the best for you."

There are solutions where you can include the best of both worlds (discrete and integrated). Goyal adds, "A system having both integrated and discrete GPUs switches intelligently between the two depending on what application you are running. So you can enjoy an unprecedented combination of power efficiency and performance."

From a designer's perspective, "Discrete GPUs are good for highly accelerated software, where a certain processing block can be completely off-loaded to the GPU. In the case of a software needing continuous data transfer between a host and the GPU, an integrated GPU would perform better than a discrete GPU, as there is a lot of data movement across GPU and CPU. So it really depends on the software—how is it designed and will it perform better on integrated GPUs than discrete GPUs," shares Kapil Agrawal, designer and CEO, Mediamaic Technologies.

Bhandari adds, "You will see almost five times improvement in data transfer when the GPU is closer to the CPU as in the case of an integrated platform."

Whatever the needs of end users, it's clear that the right GPU—whether it's discrete or integrated—is the key to a great computing experience. As almost every application becomes graphically oriented, it's the GPU in your system which will ensure that you get the best bang for your buck in a device.

Graphics on CPU silicon

"Earlier, integrated GPUs meant GPU and CPU as separate pieces of silicon on the same motherboard. With technological advancements, integrated GPUs have started including graphics onto the same die as the processor," shares Bhandari.

Bringing graphics and processor core onto the same die has a lot of advantages, explains Bhandari. There